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A comprehensive evaluation of large language models for information extraction from unstructured electronic health records in residential aged care

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ABSTRACT

Despite rapid healthcare digitization, extracting information from unstructured electronic health records (EHRs), such as nursing notes, remains challenging due to inconsistencies and ambiguities in clinical documentation. Generative large language models (LLMs) have emerged as promising tools for automating information extraction (IE); however, their application in real-world clinical settings, such as residential aged care (RAC), is limited by critical gaps. Prior studies have often focused on structured EHR data and conventional evaluation metrics such as accuracy and F1 score, overlooking critical aspects like robustness, fairness, bias, and contextual relevance, particularly in unstructured clinical narratives. To address these gaps, this study develops a holistic evaluation framework for clinical IE from free-text nursing notes in the Australian RAC. We systematically evaluate 17 LLMs, including general-purpose and healthcare-specific variants (e.g., LLaMA, Mistral, Gemini, T5) across retrieval-augmented generation (RAG) frameworks and few-shot learning configurations (one-shot, three-shot, four-shot, five-shot). The evaluation focuses on two clinical IE tasks: named entity recognition (NER) and summarization. Results reveal LLaMA 3.1 achieved 88.58 % accuracy, 87.43 % F1 score in NER, 88.18 % F1 score, and 83.15 % relevance in summarization. However, robustness remained low (4.00 % for NER, 4.31 % for summarization) despite excellent fairness (99.9 %) and minimal bias (0.11 %) in both tasks. Further, healthcare-specific LLMs slightly outperform general models, and RAG-based approaches (LangChain, LlamaIndex) yield superior results. Task-specific optimal few-shot settings emerged: three-shot for NER and five-shot for summarization. This study provides a foundation for safely integrating generative AI into clinical decision support.

1. Introduction

The volume of data within electronic health records (EHRs) has grown exponentially in recent years. While some sections of EHRs are structured, the majority, accounting for up to 80 % of the content, remains unstructured [1]. These unstructured data contain rich clinical details, including patient conditions, diagnostic results, treatments, and outcomes that are critical for informed clinical decision-making and effective communication among healthcare professionals [2,3]. However, the manual navigation and interpretation of this text remains one of the most labor-intensive aspects of clinical workflows, often delaying the retrieval of critical information and potentially compromising patient care quality and safety [1,4]. Without powerful tools to unlock, retrieve, and process the free-text notes, healthcare providers often struggle to access critical information promptly, which can negatively affect patient care quality and safety. Extracting actionable insights from unstructured EHR data is a complex task [5]. Traditional information extraction (IE) systems, which rely on dictionary- and rule-based methods, often fall short when applied to clinical narratives. These narratives are rich with domain-specific language, abbreviations, spelling variations, and nuanced context that such systems struggle to process effectively [1,6,7]. While transformer-based models, particularly bidirectional encoder representations from transformers (BERT), have demonstrated notable improvements in clinical IE, significant challenges persist due to the inherent variability and complexity of clinical documentation practices [8]. BERT-based approaches face two

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primary limitations in clinical contexts: (1) substantial training data requirements that may exceed available annotated clinical corpora [2], and (2) sensitivity to the linguistic heterogeneity characteristic of clinical notes, including domain-specific terminology, abbreviations, and non-standardized documentation formats [8]. These constraints highlight the need for more robust IE methodologies that can effectively handle the semantic ambiguity and syntactic variation prevalent in clinical language while maintaining performance under limited training data conditions.

Recent advances in large language models (LLMs) powered by generative artificial intelligence (AI), such as GPT, T5, and LLaMA, offer promising alternatives for clinical IE. These models have demonstrated an impressive capacity for understanding and generating natural language, enabling them to perform IE tasks with minimal fine-tuning [8, 9]. However, these LLMs cannot simply be applied out-of-the-box; fine-tuning or task-specific adaptation is usually necessary to achieve optimal performance in specialized domains like clinical IE [8]. A recent trend to further improve performance and minimize the need for extensive adaptation of LLMs is the use of retrieval-augmented generation (RAG) [8,9]. RAG combines external knowledge retrieval with generation capabilities, allowing models to access relevant information dynamically during inference [8]. Along with this, prompt engineering techniques such as zero-shot and few-shot prompting enable LLMs to generalize effectively from very limited examples. Together, these approaches have gained increasing interest in enhancing the accuracy and efficiency of clinical IE within healthcare settings [8,9].

However, fine-tuning generative LLMs alone is not sufficient to ensure trustworthy and effective clinical IE [10]. To address this limitation, a holistic evaluation framework is necessary to comprehensively assess LLM performance across multiple dimensions. Key evaluation metrics include accuracy, F1 score, robustness, fairness, bias, and relevance. Accuracy reflects the reliability of the generated information [11, 12], while the F1 score captures the harmonic mean of precision and recall [12,13]. Robustness evaluates the model's stability when subjected to input perturbations or noise [14]. Fairness examines disparities in model outputs across different social or demographic groups, aiming to identify potential inequities [15]. Bias captures systematic tendencies in a model's responses that arise from skewed training data or architectural constraints [15,16]. Finally, relevance assesses the degree to which the model's output semantically aligns with the input query, prompt, or context [15,16].

Despite these advancements, two critical gaps remain that hinder the effective deployment of LLMs in real-world clinical settings such as Australian residential aged care (RAC). First, there is an absence of evaluation frameworks tailored to the nursing progress notes commonly used in RAC. These notes consist of free-text entries that document residents' daily activities, clinical observations, assessments of care needs (including risk factors), and nursing interventions [3]. As such, they are rich in context and updated frequently [9]; however, their free-text nature also makes them highly fragmented and inconsistently structured [2].

A recent review found that only 5 % of the 519 studies on LLM applications in healthcare (evaluated between January 1 and February 19, 2024) utilized actual patient care data [17]. Among the limited studies using real clinical data, most evaluation frameworks are designed for structured, coded inputs or synthetic datasets [18], which fail to capture the complexity and variability inherent in real nursing documentation. For example, while studies such as Singhal et al. [19] and Bolton et al. [20] assessed LLMs using real data, their focus remained on structured formats for tasks like question answering, overlooking the nuanced and context-rich nature of nursing notes. To the best of our knowledge, no prior work has specifically addressed the unique challenges of applying LLMs to nursing progress notes in RAC. This omission is significant, as LLMs trained primarily on clean or synthetic data may struggle to process the noisy and heterogeneous nature of real nursing notes. In such conditions, robustness is essential. Furthermore, LLMs trained on generalized public datasets frequently lack representation of the linguistic and demographic diversity present in RAC. For instance, a model trained predominantly in U.S. English may perform poorly when interpreting nursing notes from Australian RAC due to differences in terminology, phrasing, and clinical context. Such mismatches can lead to outputs that are inaccurate, biased, unfair, or contextually irrelevant. Addressing this gap is important not only for ensuring effective use of RAC data in its context but also for improving generalizability across diverse healthcare environments, since the majority of their EHR data is in unstructured formats [1].

Second, there is a scarcity of systematic comparisons across different generative LLMs, including both general-purpose and domain-specific models, as well as among training techniques such as RAG. Without such comparative analyses, it remains unclear which models and RAG configurations are best suited for diverse IE tasks in RAC settings [8]. Notably, none of the existing studies provides a comparison of these model types and RAG configurations under a holistic evaluation framework that incorporates metrics such as robustness, fairness, and contextual relevance, further limiting our understanding of their practical suitability.

To address these gaps, this study proposes two key contributions: (1) the development of a holistic evaluation framework to assess LLM performance on real-world nursing notes, emphasizing robustness, fairness, bias, and contextual relevance for clinical IE in RAC; and (2) a systematic comparison of multiple LLMs and RAG configurations to identify optimal strategies for this task. By grounding both evaluation and comparison in authentic, real-world clinical narratives, this research aims to facilitate the reliable, equitable, and context-aware deployment of generative LLMs in the underexplored but critical setting of Australian RAC.

2. Related work

2.1. Types and causes of errors in large language models

Despite their advanced generative capabilities, LLMs are prone to errors due to their reliance on statistical patterns rather than actual reasoning or understanding [21]. Based on prior studies examining LLM-generated outputs in clinical contexts, commonly observed errors include relevance errors, fabricated information, false negatives, and repetitions or redundancies [9,22–24].

2.1.1. Relevance errors

Large language models can produce responses that are irrelevant to the prompt [25]. For instance, when asked about dementia symptoms, a model might mistakenly include unrelated symptoms, such as those associated with frailty, instead of dementia. Tang et al. [23] reported relevance errors in their evaluation of LLMs on clinical evidence summarization tasks.

2.1.2. Fabricated information

Fabricated information (often misleadingly referred to as "hallucinations") occurs when LLMs generate content that is entirely fabricated or misleading [23]. In a study on clinical IE using nursing notes in RAC, Alkhalaf et al. [9] identified two types of fabricated information generated by LLMs: (1) intrinsic fabrications: the generated content is based on information present in the input data but is misrepresented or synthesized incorrectly, and (2) extrinsic fabrications: the generated content is not supported by the source material at all [9].

2.1.3. False negative

False negative occurs when LLMs fail to include critical or relevant information in their responses. For example, in a clinical note summarization task, an LLM might leave out key symptoms or treatment details. This issue has also been observed in related studies; for example, Zhang et al. [24] reported similar false negatives, highlighting that LLMs often struggle to capture all clinically important information.

2.1.4. Repetition and redundancy

Large language models may generate repetitive or redundant text, which can compromise the coherence and overall quality of their outputs [26]. For example, Searle et al. [22] reported redundancy in LLM-generated clinical text, emphasizing the importance of controlling verbosity to maintain the usefulness of generated output.

2.2. Metrics and methods for assessing large language model performance

Various metrics such as accuracy, F1 score, robustness, fairness, bias, and relevance have been used with different methods to evaluate LLMs' performance.

2.2.1. Accuracy

Accuracy is commonly assessed using three distinct metrics: exact match, perplexity, and statistical accuracy. Exact match measures the percentage of model outputs that exactly match the reference answers [11,12]. Perplexity evaluates how well the model predicts a sequence of text, with lower values indicating better predictive performance [27]. Statistical accuracy refers to the proportion of correct predictions made by the model across a dataset [28].

2.2.2. F1 score

F1 score can be computed in two main ways: token-level F1 and embedding-based F1 [12,13]. Token-level F1 measures the exact token overlap between model outputs and reference texts [12]. However, this approach is often insufficient for evaluating outputs from LLMs, particularly in healthcare, where semantic accuracy and contextual understanding are critical. As a solution, embedding-based F1, such as that used in BERTScore, measures semantic similarity using contextualized word embeddings [13]. This method maps predicted tokens to their most semantically similar counterparts in reference, capturing paraphrasing and nuanced meaning.

2.2.3. Robustness

Robustness metrics assess model stability under perturbations, typically through: (1) input transformations (e.g., noise injection, word swaps); (2) analysis of internal model behavior (e.g., weights, gradients); and (3) benchmarking on diverse, challenging datasets [29,30]. However, standard benchmarks often overlook the unique noise present in real-world clinical text, such as misspellings, abbreviations, and jargon [31]. Robustness to such low-quality input is especially important in healthcare, where models must perform reliably despite messy or inconsistent data.

2.2.4. Fairness

Fairness can be evaluated in two main ways: counterfactual fairness and statistical fairness [15]. Counterfactual fairness assesses whether a model's prediction for an individual remains the same in a counterfactual scenario where only the protected attribute (e.g., gender or race) is changed. In contrast, statistical fairness assesses whether specific performance metrics are balanced across protected groups [32]. These methods also serve as practical means of identifying and quantifying bias, which typically arises when the model exhibits systematic disparities in performance [33,34]. An example is temporal bias, which occurs when a model's performance varies significantly over time or across different temporal cohorts [35].

2.2.5. Relevance

Unlike classification tasks, text generation tasks such as summarization or translation cannot be effectively evaluated using exact match methods. This is because the outputs often involve paraphrasing, reordering, or using different word variations while still preserving the original meaning [13,36,37]. Therefore, relevance is typically measured using a range of automated metrics, including ROUGE and BERTScore, as well as others like BLEU, BLEURT, and METEOR [13,36–38]. These metrics assess different aspects of semantic similarity and surface-level overlap, making them suitable for evaluating the quality and contextual alignment of generated text.

While these metrics are considered essential for holistic evaluations, their applicability and interpretability, particularly in the RAC context, remain under exploration. This raises an important research question:

> **RQ1:** To what extent do holistic performance metrics, such as robustness, fairness, bias, and contextual relevance, capture insights into the performance of large language models across clinical information extraction tasks?

2.3. Large language models for health applications

With appropriate prompting and task design, high-performing models like OpenAI's GPT-4o, Google's Gemini, and PaLM can achieve competitive results in healthcare NLP tasks [39,40]. However, their reliance on public APIs and remote processing environments often limits their suitability for healthcare applications, where data must remain within secure, private infrastructures. Healthcare organizations face ethical and legal challenges in adapting these proprietary LLMs, such as licensing agreements that may conflict with specific legal requirements for stringent personal data governance regulations in Australia. This misalignment has led healthcare organizations to increasingly consider open-source LLMs, which provide greater flexibility in deployment and customization, allowing for adherence to local data governance standards.

Open-source LLMs can be broadly categorized into general-purpose and domain-specific models, such as those tailored for healthcare [41]. General-purpose LLMs such as GPT, LLaMA, T5, and Mistral are designed to perform a wide range of tasks across multiple domains, offering flexibility and scalability. However, their application in clinical settings is often limited by a lack of specialized training on health-specific data [42]. Conversely, health-specific LLMs like Me-LLaMA, Clinical-T5, BioGPT, and Med-PaLM are trained on curated biomedical corpora, including scientific literature, clinical guidelines, and EHRs [42–45]. Thus, they are optimized for clinical tasks like question answering, named entity recognition (NER), relation extraction, and summarization. For instance, Me-LLaMA has demonstrated superior performance compared to general-purpose models like LLaMA across several benchmark clinical NLP tasks [42].

Despite their domain specialization, health-specific LLMs are not without limitations. Most are trained primarily on structured clinical data, with little to no exposure to unstructured free-text formats such as nursing progress notes. Moreover, the training data for these models is predominantly U.S.-based, raising concerns about their applicability to Australian healthcare contexts. Thus, a significant knowledge gap remains in determining the optimal choice of open-source models, whether general-purpose or health-specific, for tasks such as mining free-text nursing progress notes within the Australian RAC context. This leads to the research question:

> **RQ2**: Which type of LLM, general-purpose or health-specific, performs better in clinical information extraction tasks when evaluated using holistic metrics like robustness, fairness, and contextual relevance?

2.4. Retrieval-augmented generation

RAG is a method that enhances the accuracy and relevance of LLMs by retrieving task-specific external knowledge, rather than relying solely on internal training data [8,9]. This approach is especially valuable in clinical IE, where task-specific or domain-specific context is essential. The external knowledge sources used in RAG are expected to contain relevant and high-quality information tailored to the task [9]. However, if the content within the knowledge source is irrelevant or low quality, the model may produce inaccurate or fabricated outputs, reducing the reliability of the generated information [46]. Further, to manage this retrieval process, RAG depends on retrieval frameworks that handle the connection between the user query and the knowledge source [9,47]. Two widely used RAG frameworks are LlamaIndex and LangChain, both of which provide tools to efficiently organize and retrieve external data [47]. These frameworks are responsible for searching, ranking, and retrieving the most relevant documents to be passed on to the LLM.

In a RAG approach, users typically submit a query related to a specific task, such as clinical IE, through a process known as prompting [8]. This step provides the model with sufficient context to understand the query and generate a relevant response. For example, few-shot prompting embeds a small set of exemplar question-answer pairs within the prompt to guide the model's reasoning [48]. When combined with retrieved evidence, this method can significantly enhance the relevance and accuracy of the output. However, the effectiveness of prompting can be influenced by prompt sensitivity, meaning that variations in the prompt's structure or wording may affect the quality of the response [49]. In such cases, both the format of the prompt and the number of examples included must be carefully optimized to ensure accurate and context-aware results [49,50]. Therefore, both the selection of the retrieval framework and the engineering of prompts jointly determine the effectiveness of RAG systems for clinical IE. Yet, a key gap remains in determining the optimal RAG configuration, specifically, the choice of framework (e.g., LlamaIndex vs. LangChain) and the ideal number of few-shot examples for clinical IE from free-text nursing progress notes in the Australian RAC setting. This leads to the following research questions:

RQ3: Which RAG framework delivers the best performance in clinical information extraction tasks when evaluated using holistic metrics like robustness, fairness, and contextual relevance?
RQ4: What is the optimal number of examples in few-shot learning with RAG to maximize performance in clinical information extraction tasks when evaluated using holistic metrics like robustness, fairness, and contextual relevance?

This work addresses these research questions through a comprehensive evaluation framework and comparative analysis, as outlined in the following sections.

3. Methodology

The methodology comprises four key steps: (1) identifying the essential experimental components, including the dataset selection, scenario definitions, LLM models, prompt structures, LLM training methods, and evaluation metrics; (2) conducting the experiment; (3) applying the evaluation framework; and (4) performing statistical analyses to compare and interpret the evaluation results.

3.1. Step 1: experimental components definition

Ethics approval

The Human Research Ethics Committee of the University of Wollongong approved the study (Ethics Number 2019/159).

3.1.1. Dataset selection

De-identified demographic data and two million free-text electronic nursing progress notes were collected from 40 RAC facilities belonging to one management group in New South Wales, Australia, from 2019 to 2021. During text preprocessing, non-textual characters, such as extra delimiters and blank spaces, were removed. However, we retained a range of stop words, e.g., “a", “be,” “very,” “should,” for their semantic relevance to the clinical context [51].

3.1.2. Scenario definition

A scenario represents a specific use case for an LLM, defining the tasks we want the models to perform. We developed a structured approach (see Fig. 1), breaking down each scenario into key components: the task and the domain, inspired by previous research [15]. The tasks include NER and summarization (see Table 1). The NER task targets clinically relevant information such as symptoms, treatments, and outcomes across care recipient records [8]. Summarization, in contrast, condenses lengthy care recipient histories and clinical notes into concise, actionable insights, enabling healthcare professionals to make timely decisions, particularly in high-demand environments like RAC [9,52]. Therefore, our selection of clinical use cases aligns with the development direction of LLMs for healthcare and maximizes impact.

The domain is categorized into three key aspects: the content of the text (what), the speaker or source of the text (who), and the temporal context of the text (when). These classifications provide a structured framework for analyzing free-text nursing progress notes in RAC. The “what” (text content) captures the subject matter and stylistic variations in the text. Examples include documentation of clinical contexts such as agitation in dementia and malnutrition diagnosis. These categories reflect the specific health and care concerns commonly documented in RAC.

The “who” refers to a specific male or female resident, identified by a pseudo-code for de-identification to protect privacy. The “when” (temporal context) denotes the period when the text (nursing notes) was generated, such as 2019, 2020, or 2021. This structured scenario framework was carefully designed with the following considerations: (1) information documented in free-text nursing progress notes is the primary data source in RAC; (2) the framework aligns with aged care information requirements, ensuring relevance and applicability to the domain; and (3) the research team has curated an annotated dataset for training and testing models, enabling robust and reliable performance evaluation. The annotated dataset for agitation in dementia was derived from the studies by Zhu et al. [53,54], and the dataset for malnutrition was based on the work of Alkhalaf et al. [55]. Clinical domains (referred to as 'what' in the structured scenario framework) consist of varying numbers of clinical labels, with counts ranging from 13 to 85, as shown in Table 2.

3.1.3. Large language models

We evaluated eight general-purpose LLMs and nine health-specific LLMs, as detailed in Table 3. Model selection is based on several key considerations: (1) addressing practical challenges related to GPU resource availability; (2) ensuring ease of deployment, convenience, and operational control over usage on local servers; (3) compliance with Australian health data privacy laws; (4) widespread adoption and community support; and (5) demonstrating strong performance across a range of tasks.

3.1.4. Prompt structures

The final prompts used in our experiments are listed in Supplementary Table 1 for NER tasks and Supplementary Table 2 for summarization tasks, as inspired by previous studies [8,9].

3.1.5. LLM training method

We adopted the training strategy of RAG with few-shot learning as recommended by Vithanage et al. [8]. We applied RAG frameworks: LangChain and LlamaIndex, and one-shot, three-shot, four-shot, and five-shot configurations as few-shot learning settings. To implement few-shot prompting, we manually constructed prompts using representative examples selected from labeled datasets. These labeled instances were sourced from previously published studies: Zhu et al. [53,54] for agitation in dementia, and Alkhalaf et al. [55] for malnutrition. All examples were independent of the RAG knowledge base and test dataset. The initial selection was random. Recognizing the sensitivity of few-shot performance to both the phrasing and order of examples, we

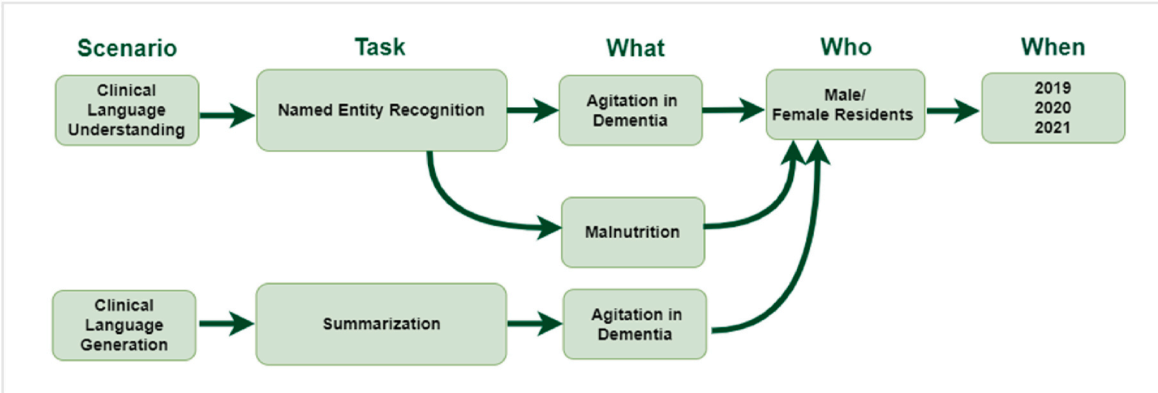

Fig. 1. The structured scenario framework, including clinical tasks, the "what", "who", and "when" context.

Table 1
Examples of NER and summarization tasks.

Task	Example
Named entity recognition	Resident requires comprehensive assistance with his hygiene and toileting needs due to his confusion. A nurse is responsible for assisting him with toileting and changing his pads. Malnutrition risk factors: confusion
Summarization	The resident displayed symptoms of agitation, including hitting another resident in the dining area. Staff identified loud noises as a potential trigger for this behaviour. The two residents were immediately separated and will remain apart for the rest of the day. After the separation, the resident settled down. Summary Agitation symptoms: agitation, hitting Triggers: loud noises Interventions: separation of the two residents Location: dining area Outcome: settle

standardized the prompt templates by fixing the wording and sequence of examples. These prompt templates were held constant across all models and experimental runs to minimize variability and control for randomness in evaluation. A sample of a few-shot examples used in the study is in Table 4.

3.1.6. Evaluation metrics

The selection of evaluation metrics, accuracy, F1 score, robustness, fairness, bias, and relevance, was inspired by prior studies [15]. The experimental framework is shown in Table 5.

3.2. Step 2: experimentation

To execute the experiment, we specified a series of runs, and each run is defined by a six-element tuple derived from Step 1: data, scenario, LLM, prompt, training method, and evaluation metric, as below.

$$\text{Experiment} = \{\text{Run\_}(1), \text{Run\_}(2), \ldots, \text{Run \_}(n)$$

$$\text{Run}_i = (\text{Data}_i, \text{Scenario}_i, \text{LLM}_i, \text{Prompt}_i, \text{Training Method}_i, \text{Evaluation Metric}_i)$$

In each run, the LLM is augmented with retrieval using the RAG framework. The scenario defines the task to be performed, mainly driven by the query provided through the prompt. The input data (test dataset) is then processed by the LLM, which generates output responses. Finally, these outputs are evaluated using the chosen evaluation metrics to measure the model's performance. This overall process is illustrated in the formula below:

$$(\text{Data}_i, \text{Scenario}_i, \text{Prompt}_i, \text{Training Method}_i) \rightarrow \text{LLM}_i \rightarrow \text{Metric}_i(\text{LLM}_i \text{ Output})$$

The workflow of RAG is illustrated in Algorithm 1 [9]. The experiment was conducted on four NVIDIA RTX-A5000, each equipped with 32 GB of memory.

The software environment was Ubuntu 18.04, Python 3.12.0, and PyTorch 2.2. The experiment began with the preparation of domain-specific textual data (see Table 6), which was essential for building a tailored knowledge base required by the LLMs during the RAG process. To achieve this, we utilized the annotated data described in Section 3.1.3. From the annotated data, we filtered records from 2019 for RAG file preparation. A sample of data for RAG is presented in Table 7.

This data was formatted in JSON and input into an embedding model, which generated embeddings that were subsequently stored in a vector store. To perform the task of NER and summarization, we formulated a query by combining the prompt with few-shot learning and test data. The retriever then used this query to retrieve relevant notes from the vector store, which were fed to the LLM to generate the final output (see Supplementary Tables 1 and 2 for sample outputs). Large language models were quantized throughout the experiment to reduce their memory footprint, enhancing efficiency and suitability for deployment in real-world applications. For this experiment, the RAG frameworks were implemented in three stages: first, applying LangChain alone, then LlamaIndex, and finally combining both frameworks to assess performance improvements. In the combined setup, LangChain and LlamaIndex were used sequentially. Specifically, LlamaIndex handled the core document indexing and retrieval pipeline using a VectorStoreIndex, while LangChain was incorporated via its HuggingFaceEmbeddings wrapper, which was passed into LlamaIndex for generating embeddings.

For this study, we utilized a test dataset (see Table 8) comprising nursing notes (excluding those included in the RAG knowledge source), selected from the annotated dataset with ground truth labels (see Section 3.1.3). Testing was conducted over a period of three weeks (n = 3). To account for potential variability due to model randomness in generative AI, multiple runs were performed under consistent conditions, ensuring reliable and reproducible results. Additionally, to prevent any residual effects from previous runs, we reloaded the LLMs from the HuggingFace library at the beginning of each session for both tasks, maintaining consistent initial conditions. To further quantify uncertainty and enhance evaluation robustness, we applied bootstrapping by resampling the performance scores from the three runs 1000 times with replacement, following prior work [67,68]. This helped stabilize performance and variability estimates, particularly given the limited number of runs.

Table 2
Name and number of clinical labels in each domain.

Clinical Domain	Clinical Labels	Number of Clinical Labels
Agitation in dementia	**Symptoms:** disruptive vocalisation, verbally aggressive behaviour, arguing, complaining, cursing, threat, using abusive language, using accusatory language, using foul language, using hostile language, using obscene language, using profane language, verbally nonaggressive behaviour, ceaseless talking, constant repetition of word, constant unwarranted requests for attention, constant unwarranted requests for help, constant unwarranted requests for reassurance, echolalia, groaning, grunting, howling, making bizarre noise, rambling, repetitive questioning, roaring, screaming, shouting, speaking in excessively loud voice, emotional distress, anger, frustration, irritability, mood swing, negativism, outburst, physically aggressive behaviour, biting, destroying property, fighting, grabbing, hitting, hurting self, hurting someone, kicking, pushing, resisting, scratching, shoving, slamming, spitting on people, staring, striking people, tearing, throwing object, physically nonaggressive behaviour, constant manipulation of object, fidgeting, frightening others, gesturing, hand wringing, inappropriate dressing, inappropriate handling object, inappropriate undressing, pacing, pointing finger, repetitive physical mannerism, restlessness, rocking, rummaging, searching, wandering, bruxism, resisting, punching, absconding, calling out, physical agitation, facial grimacing, paranoia, moving objects, hoard items, intrusive of others privacy, gets up and down from constantly, urinating on the floor.	85
	Triggers: tired, infections, too hot, too cold, unfamiliar tasks, unmet hygiene needs, pain, loud noise, discomfort, constipation, hunger, unfamiliar staff and activity, thirst, obsession, compulsive disorder, unable to communicate, scare, other people's aggressive behavior, boredom, forgetful of time and place, frustration, loneliness, confusion, miscommunication, low mood, urinary urgency, episodes of incontinence, poor insight, distress, cognitive impairment, anxious, lack of ability to express the needs, lack of understanding of care needs, sundowning, change in routine, crowded environment, inappropriate routine, lack of appropriate engaging activity, phone call frequency, social isolation, speaking the local language as a second language, visitor visit frequency, intrusion into personal privacy, intrusion into personal space, visitor behavior, thirst, depression, medication change.	48
	Interventions: redirection, separating from other residents, reassurance, encouragement, checking regularly, monitoring, providing assistance, approaching gently, spending time, distraction, music therapy, playing cards, doing simple puzzles, reorientating to time and place, speaking gently, diversional therapy, PRN dose, verbal prompting, relocate, positive reinforcements, constant supervision, causative factor investigation, family member focused intervention, having regular staff, psychological treatment.	25
Malnutrition	**Risk factors:** anxiety, bowel blockage, cancer, chest infection, chronic wound, confusion, constipation, delirium, dementia, depression, diabetes, diarrhea, difficulty swallowing, dysphagia, eating disorder, food preference, frailty, gastritis, heart disease, blood glucose, hospital admission, isolation, refusal of care, fall, food modification, nausea, Parkinson, pneumonia, poor appetite, poor intake, poor oral health, pressure ulcer, sepsis, stroke, fluid modification, surgery, vomiting.	37

Algorithm 1. RAG inference process

Input: RAG reference file $\boldsymbol{D_{RAG}}$, standard test dataset **D,** embedding model ***EM***, vector store ***VS***, prompt ***P***, large language model ***LLM***, few-shot examples ***E***, metrics **M**
Note: i denotes the current iteration during the process and k indicates the individual few-shot example.

1. Embeddings $= \mathrm{EM}(\mathrm{D}_{RAG})$//Generate embeddings using the embedding model
2. VS $= \mathrm{Store}(\mathrm{Embeddings})$//Store embeddings in the vector store
3. $\mathrm{Query}_i = \mathrm{P} + \sum_{k+1}^{n} \mathrm{E}_{k,i} + \mathrm{D}$//Construct the prompt with a few-shot example
//Where n represents the number of examples (0, 1, 3, 4, 5)

4. Retrieved $\mathrm{Notes}_i = \mathrm{Retrieve}(\mathrm{Query}_i, VS)$//Fetch relevant notes from the vector store using
the query

5. $\mathrm{Output}_i = \mathrm{LLM}(\mathrm{Query}_i, \text{Retrieved Notes}_i)$//Generate a response using the LLM
6. $\mathrm{Response}_i = \mathrm{Evaluate}(\mathrm{Output}_i, \mathrm{M})$//Evaluate the response using specific metrics

Output: Response//Responses from RAG

*//denotes comments explaining each step of the algorithm. The same notation applies to the algorithms that follow.

3.3. Step 3: model performance evaluation

The outputs from the LLMs, generated in JSON format, were evaluated against annotated ground truth data, also in JSON format, for both NER and summarization tasks. Details of the test data used in the experiment are provided in Table 8.

3.3.1. F1 score

We calculated embedding-based F1 using BERTScore, following the evaluation methodologies outlined by Li et al. [64] and Vithanage et al.

Table 3
LLMs used in the study.

Type of LLMs	LLMs	Baseline LLM	Number of Parameters
General LLMs	LLaMA-2-Chat-7B [56]	–	7 billion (7B)
	LLaMA-2-Chat-13B [56]	–	13B
	LLaMA-3-8B [57]	–	8B
	LLaMA3.1–8B [57]	–	8B
	Mistral [58]	–	7B
	Gemini [59]	–	13B
	Qwen [60]	–	8B
	T5-Base [61]	–	220 million (M)
Health-specific LLMs	MedAlpaca-7B [62]	LLaMA-7B	7B
	MedAlpaca-13B [62]	LLaMA-13B	13B
	Baize-Healthcare [63]	LLaMA-7B	7B
	ChatDoctor [64]	LLaMA-7B	7B
	Asclepius-7B [65]	LLaMA-2-7B	7B
	Asclepius-13B [65]	LLaMA-2-13B	13B
	PMC LLaMA [66]	LLaMA-2-13B	13B
	Me-LLaMA [42]	LLaMA-2-13B	13B
	Clinical-T5-Base [44]	T5-Base	220M

Table 4
Sample of a few-shot examples used in the study.

Task	Few-shot Examples
Named entity recognition	Example 1: Nursing note: "Resident, who suffers from vascular dementia, displays notable physical agitation/aggression and verbal behaviours. The resident exhibits verbal disruption, calling out, and screaming, which disturbs others. The resident frequently requires staff assistance for reassurance, comfort, and distraction using strategies such as music therapy, playing cards, or engaging in simple puzzles." Identified symptoms: physical agitation/aggression, verbal disruption, calling out, screaming. Example 2: Nursing note: "Resident requires the assistance of a nurse for his hygiene and toileting needs. Additionally, he is prone to constipation and receives aperients as necessary. The current intervention measures in place have proven to be effective." Identified risk factor: confusion
Summarization	Example 1: Nursing note: Staff reported that the resident exhibited physically aggressive behaviour towards another resident, scratching him and causing a skin tear in the dining room. In response, staff separated the residents and closely monitored residents' behaviour. They were settled after these interventions." Agitation in Dementia Summary: Symptoms: [physically aggressive behaviour: scratching] Triggers: [ ] Interventions: [separating the residents involved, close monitoring] Location: [dining room] Outcomes: [settled]

[8].

3.3.2. Robustness

To evaluate robustness, we calculated a robustness score (see Algorithm 2) inspired by the previous studies [14]. This involved introducing different types of perturbations to the input nursing notes and measuring the F1 score to assess the stability of the model's performance. These perturbations included: (1) applying upper case transformations, (2) introducing common typos (e.g., "pain" becomes "painn" or "medication" becomes "medicaiton."), (3) adding random numbers, (4) using abbreviations e.g., "blood pressure" becomes "BP" and "heart rate" becomes "HR"), (5) injecting slang terms (e.g., "medication" might become "meds"), and (6) replacing adjectives with synonyms (e.g., "severe" becomes "intense" and "moderate" becomes "average").

Algorithm 2. Robustness evaluation metric inference process

Input: RAG reference file $\boldsymbol{D_{RAG}}$, standard test dataset $\boldsymbol{D}$, noisy test dataset $D_{\_noise}$

1. $M = \text{Train\_Model}(D_{RAG}, \text{Method} = \text{Few} - \text{shot with RAG})$//Train Model
2. $F1_{orginal} = \text{Evaluate}(M, D)$//Calculate F1 scores
$F1_{noise} = \text{Evaluate}(M, D_{noise})$

3. $\Delta F1_{robustness} = |F1_{\text{orginal}} - F1_{\text{noise}}|$//Calculate differences in F1 scores
4. $R = (\Delta F1_{robustness})$//Calculate robustness score

Output: R//Results of robustness score

Table 6
Volume and file size of nursing notes in the RAG knowledge source for each clinical task.

Clinical Domains	Number of Nursing Records	File Size
Agitation in dementia	145	14 KB
Malnutrition	36	8 KB

3.3.3. Fairness

To evaluate fairness, we applied the counterfactual fairness method. This involved analyzing the model's performance on gender-specific input data by calculating the F1 score separately for male and female groups (refer to Algorithm 3). Additionally, we measured demographic parity by computing the absolute difference in outputs between nursing notes for male and female residents, as outlined in prior research [69].

Algorithm 3. Fairness evaluation metric inference process

Input: RAG reference file $\boldsymbol{D_{RAG}}$, test dataset for nursing notes belong to male residents $\boldsymbol{D_{male}}$, test dataset for nursing notes belonging to female residents $\mathbf{D_{female}}$

1. $M = \text{Train\_Model}(D_{RAG}, \text{Method} = \text{Few} - \text{shot with RAG})$//Train Model
2. $F1_{\text{male}} = \text{Evaluate}(M, D_{male})$//Calculate F1 scores
$F1_{\text{female}} = \text{Evaluate}(M, D_{female})$

3. $\text{Average\_Performance} = (F1_{\text{male}} + F1_{\text{female}})/2$//average performance
4. $\text{Demographic\_Parity} = |F1_{\text{male}} - F1_{\text{female}}|$//Measure of fairness

Output: $\text{Average\_Performance}, \text{Demographic\_Parity}$

3.3.4. Bias

As part of the RAG process, we input knowledge through a data file recorded in 2019 (see Section 3.2), forming the basis of the retrieval mechanism. To evaluate potential bias, we analyzed whether the model performed significantly better on the 2019 test data, which corresponds to the RAG data file, in comparison to the test data from 2020 to 2021. To evaluate the bias score (see Algorithm 4), we assessed the variance in model performance on the F1 score across test datasets from 2019, 2020, and 2021 to ensure a comprehensive assessment of the model's consistency across different periods. Absolute differences, as inspired by previous studies [70,71], were used for this evaluation.

Algorithm 4. Bias evaluation metric inference process

Input: RAG reference file $\boldsymbol{D_{RAG\text{-}2019}}$, test dataset for 2019 $\boldsymbol{D_{2019\_test}}$, test dataset for 2020 $\boldsymbol{D_{2020\_test}}$, test dataset for 2021 $\mathbf{D_{2021\_test}}$.

1. $M = \text{Train\_Model}(D_{RAG}, \text{Method} = \text{Few} - \text{shot with RAG})$//Train Model
2. $F1_{2019} = \text{Evaluate}(M, D_{2019\_test})$//Calculate F1 scores
$F1_{2020} = \text{Evaluate}(M, D_{2020\_test})$
$F1_{2021} = \text{Evaluate}(M, D_{2021\_test})$

3. $\Delta F1_{2019-2020} = |F1_{2019} - F1_{2020}|$//Calculate absolute differences
$\Delta F1_{2020-2021} = |F1_{2020} - F1_{2021}|$
$\Delta F1_{2019-2021} = |F1_{2019} - F1_{2021}|$

4. Mean Absolute Bias $= (\Delta F1_{2019-2020} + \Delta F1_{2020-2021} + \Delta F1_{2019-2021})/3$
5. Standard Deviation $= \text{StdDev}(\text{F1 scores})$

Output: Mean Absolute Bias, Standard Deviation

Table 5
Experimental framework.

Scenarios	Tasks	Clinical Domains	Accuracy	F1 score	Robustness	Fairness	Bias	Relevance
Clinical language understanding	Named entity recognition	Agitation in dementia	Y	Y	Y	Y	Y	N
		Malnutrition	Y	Y	Y	Y	Y	N
Clinical language generation	Summarization	Agitation in dementia	N	Y	Y	Y	Y	Y

Note: 'Y' denotes Yes, 'N' denotes with No.

3.3.5. Relevance

We measured relevance for the summarization task primarily using ROUGE and BERTScore, along with additional metrics such as BLEU, BLEURT, and METEOR. These metrics evaluate the similarity between the generated summaries and the reference annotated ground truth data (see Section 3.1.3). Since no single metric can fully capture all aspects of summary quality, we employed this combination of complementary metrics to provide a more comprehensive and balanced assessment of the summarization system's performance, as recommended by prior studies [72]. In particular, BLEURT and METEOR were included for their ability to capture semantic fluency and handle variations in wording, which is especially important in the healthcare domain where domain-specific synonyms and paraphrases frequently occur [73–75].

3.4. Step 4: statistical analysis

To determine the appropriate statistical tests, we first assessed the distribution of each performance metric using the Shapiro–Wilk test for normality, following prior recommendations [76]. Since the metrics were not normally distributed, we applied a logit transformation to the proportion-based scores, including accuracy, F1 score, BERTScore, ROUGE, BLEU, BLEURT, and METEOR, as suggested by previous studies [77,78]. This transformation helps stabilize variance and better approximate normality in proportion data. Post-transformation, the scores were found to approximate a normal distribution. We further assessed the assumption of homogeneity of variances using Levene's test [79]. When this assumption was met, we conducted a standard one-way analysis of variance (ANOVA) for comparisons across three or more independent groups. For comparisons between the two groups, independent samples t-tests were used. These methods are in line with recommendations from prior statistical studies [80–82]. The null hypothesis assumes no significant difference in performance metrics between groups, while the alternative posits a significant difference. A p-value below 0.05 indicates statistical significance, leading to the rejection of the null hypothesis. To control for false positives arising from multiple comparisons, we applied the Bonferroni correction, as recommended in prior studies [83].

Table 7
Sample data file for RAG.

Clinical Domains	RAG Data File	
	Record	Label
Agitation in dementia (symptoms)	The resident lacks insight into her care needs and personal safety, and refuses care most of the time.	Resistive
	The Resident repeatedly argued with peers during group activities.	Argue
	The resident hoarded multiple items in their bedside drawer.	Hoarding items
Agitation in dementia (triggers)	The resident's inability to communicate has caused increased agitation.	Unable to communicate
	Loud noise in the environment has caused agitation among the residents.	Loud noise
	The resident's boredom has contributed to their symptoms of agitation.	Boredom
Agitation in dementia (interventions)	The resident was immediately separated after hitting another resident and remained separated for the rest of the day.	Separate from other residents
	The resident often wanders and is redirected gently.	Redirection
	The resident needed continuous reassurance due to their refusal of care.	Reassurance
Malnutrition (risk factors)	The resident was given PRN Movicol for constipation and had a bowel movement this morning.	Constipation
	The resident was settled last week, except for one instance of feeling nauseous.	Nausea
	The resident had a poor appetite and minimal fluid intake.	Poor appetite

4. Results and discussions

This section presents the study's findings as responses to the research questions (RQ1, RQ2, RQ3, and RQ4) outlined in Section 2. The findings are followed by a discussion of their implications in the context of these research questions. The experimental results are detailed in Supplementary Table 3.

4.1. Answers to RQ1: to what extent do holistic performance metrics, such as robustness, fairness, bias, and contextual relevance, capture insights into the performance of large language models across clinical information extraction tasks?

To address RQ1, we employed a RAG framework that integrates LangChain and LlamaIndex, a three-shot prompting strategy for NER, and a five-shot strategy for summarization, using the LLaMA 3.1 8B model. These configurations were informed by insights from RQ3 and RQ4, as detailed in Sections 4.3 and 4.4.

4.1.1. Results

For the NER task, the model achieved an accuracy of 88.58 % (±0.02 %), an F1 score of 90.43 % (±0.02 %), a robustness score of 4.00 % (±0.37 %), a fairness score of 99.9 % (±0.04 %), and a bias score of 0.11 % (±0.02 %). For the summarization task, the F1 score was 88.18 % (±0.02 %), the robustness score of 4.31 % (±0.32 %), the fairness of 99.9 % (±0.03 %), the bias of 0.11 % (±0.03 %), and the relevance scores of 83.15 % (±0.03 %) for ROUGE, 84.45 % (±0.03 %) for BERTScore, 84.28 % (±0.04 %) for BLEURT, 83.21 % (±0.03 %) for METEOR, and 84.66 % (±0.03 %) for BLEU. These metrics collectively convey that the model performs well in terms of correctness (accuracy and F1 score), fairness, and bias. At the same time, there is some concern regarding its robustness and moderate level of relevance scores.

Further, we conducted a statistical analysis to assess robustness, fairness, and bias in the NER and summarization tasks across different conditions. For robustness assessment, we compared F1 scores in two ways: (1) between tasks (NER and summarization) using noisy data, and (2) within each task by comparing original data with noise-injected data. While no statistically significant differences were observed in either comparison, a trend suggested that original notes achieved slightly higher F1 scores in both tasks (see Fig. 2). For fairness, we compared F1 scores between male and female resident records in the original dataset in both tasks and found no significant gender difference. Similarly, for bias, we compared F1 scores across clinical notes recorded in 2019, 2020, and 2021 to detect potential temporal bias or dataset shift bias due to changes in data distribution over time. No significant differences were identified.

We manually analyzed the error notes in NER (see Supplementary

Table 8
Details of the test data used in the experiment.

Test data	Metrics	NER (N)	Summarization (N)
Original data	F1 score, accuracy, relevance	200	100
Noice data	Robustness	200	100
Male data	Fairness	200	100
Female data	Fairness	200	100
2019 data	Bias	200	100
2020 data	Bias	200	100
2021 data	Bias	200	100

Note: 'N' denotes the number of records in the test data. The same notation applies to the tables that follow.

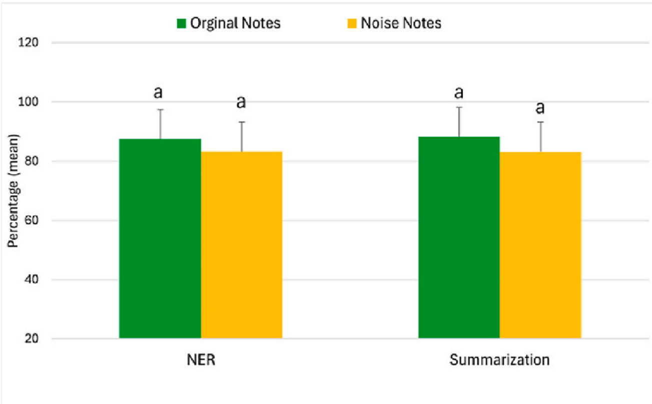

Fig. 2. Comparison of the LLMs' F1 score performance for original and noise notes. Note: Different letters above the bars denote statistically significant differences. Groups marked with the same letter are not significantly different from each other. The same notation applies to the graphs that follow.

Table 4) and summarization (see Supplementary Table 5) tasks. The error analysis (see Table 9) reveals that false negatives were the most frequent in both NER and summarization tasks for both original and noisy data, followed by intrinsic fabrications, with extrinsic fabrications being the least common. No significant repetition or redundancy errors were observed.

4.1.2. Discussion

In this study, we conducted a holistic evaluation of LLM performance for NER and summarization tasks using unstructured EHRs from real-world RAC settings. Our work addresses gaps in prior research, which has predominantly focused on conventional performance metrics such as accuracy and F1 score, while largely neglecting comprehensive assessments of robustness, fairness, bias, and clinical relevance [14]. Moreover, many previous studies have relied on publicly available or synthetic datasets or structured data, thereby limiting their applicability to actual healthcare environments [17]. By incorporating additional evaluation dimensions, our study provides a more nuanced and clinically relevant assessment of LLMs, reducing uncertainty regarding their practical deployment in healthcare.

4.1.2.1. Accuracy and F1 scores. The evaluation of accuracy revealed minimal margins of error, ±0.02 % for NER. This suggests the model's stable performance across different samples. Again, the low variability in F1 scores (±0.02 % for NER and ±0.02 % for summarization) suggests a robust balance between precision (the correctness of positive identifications) and recall (the ability to capture all relevant cases). These findings collectively demonstrate that the model consistently identifies relevant elements within the text, thereby reinforcing the reliability of these traditional performance metrics in our context.

Table 9
Error prevalence in the experiment.

Task	Error Type	Original Data (%)	Noisy Data (%)
NER (N = 200)	False negative	12 %	18 %
	Intrinsic fabrications	8 %	8 %
	Extrinsic fabrications	4 %	6 %
	Repetition or redundancy	0 %	0 %
Summarization (N = 100)	False negative	8 %	13 %
	Intrinsic fabrications	4 %	2 %
	Extrinsic fabrications	2 %	2 %
	Repetition or redundancy	0 %	0 %
Total Error Rate		38 %	49 %

4.1.2.2. Robustness. Robustness is critical for real-world applications because many NLP systems exhibit sensitivity to minor input perturbations, which can compromise their generalizability. In our study, robustness was evaluated by applying controlled noise to the input text (see Supplementary Tables 4 and 5 for sample noise data). Previous studies reported robustness scores of 17 % for LLaMA 2 [14] and 17 % using similar perturbation methods [14]. In contrast, our evaluation yielded lower robustness scores (4.00 % for NER and 4.31 % for summarization), which may reflect differences arising from our use of LLaMA 3, a more advanced model that might inherently respond differently to noise.

Despite the lower scores, the relatively high standard deviations (±0.32–0.37 %) indicated substantial variability in model performance. Additionally, while not statistically significant, original notes tended to outperform noisy ones in F1 scores, suggesting the model's sensitivity to input variations. It is important to note that our robustness evaluation was limited to textual noise. We did not investigate additional forms of perturbation, such as weight, gradient, or activation noise [84,85]. Future research incorporating these dimensions could provide a more comprehensive assessment of model robustness.

4.1.2.3. Fairness. Fairness in our study was assessed by measuring performance equity across male and female resident records using the fairness computation defined in Algorithm 3. Our model demonstrated no significant difference in F1 scores between male and female resident records and a near-perfect fairness score of 99.9 % for both NER and summarization with negligible variability (±0.03–0.04 %), indicating that performance is consistent across these groups. Our performance is much better than the 58.5 % achieved in previous research [89] with the LLaMA 3 model, focusing on the race-targeted question and answering tasks. Such a high fairness score is feasible in our case because the calculation was limited to gender as the only demographic variable, which we acknowledge as a limitation. While socioeconomic status and race [86] were less relevant in our dataset, given the uniformity of care services in Australian RAC facilities, future studies could incorporate additional demographic factors, such as age and special care needs, to capture a broader spectrum of potential disparities.

4.1.2.4. Bias. Bias was quantified by comparing model performance across clinical notes from different years (2019, 2020, and 2021), collected from 40 RAC facilities, with the training dataset predominantly originating from 2019, as defined in Algorithm 4. Our results indicate a notably low bias score of 0.11 %, with minimal variation (±0.02–0.03 %) and no significant difference in F1 scores across the temporal cohorts. While there is no universally defined threshold for acceptable bias in longitudinal model evaluation, scores close to 0 are generally considered negligible bias in prior studies [87,88]. These findings suggest that the model exhibits minimal temporal bias within the RAC context. Nonetheless, our approach does not account for other potential sources of bias, such as variations in clinical terminology, cultural contexts, or institutional practices. This limitation may have contributed to the low observed bias. Furthermore, since the knowledge source was built exclusively from 2019 data, a performance drop in 2020 and 2021 was expected. However, this decline was minimal, likely because the data were sourced from a single aged care management group under uniform policies, and inter-facility differences were likely small. We also acknowledge that the limited three-year timeframe may have constrained our ability to detect meaningful temporal shifts, as clinical definitions and documentation standards are unlikely to vary significantly over such a short period. Evaluating model performance over a longer period, such as a decade, could uncover additional biases associated with changes in treatment protocols and clinical documentation practices. While our current dataset is restricted to three years, we encourage future research to assess temporal bias more comprehensively using extended longitudinal data.

4.1.2.5. Relevance. Relevance was evaluated using ROUGE, BERTScore, BLEURT, METEOR, and BLEU to assess how well the model's outputs captured key information from the input data. Our models achieved relevance scores in the range of ~81–85 %. While there is no universally accepted threshold for interpreting these metrics, we describe this performance as moderate based on comparisons with prior studies such as Kadhim et al. [89], Afrin et al. [90], and Kieuvongngam et al. [91], which reported relevance scores around 88 % on summarization tasks. The standard deviations (±0.030 % for ROUGE, ±0.03 % for BERTScore, ±0.04 % for BLEURT, ±0.03 % for METEOR, and ±0.03 % for BLEU) further suggest consistency across runs. This suggests that while our models perform reasonably well, there remains scope for improvement in aligning model outputs more closely with key content in the source notes. However, we recognize that relying solely on these metrics may limit the depth of our evaluation. Incorporating human assessments in future work could offer deeper insight into contextual relevance and coherence.

4.1.2.6. Error analysis. A detailed examination revealed three primary types of errors: intrinsic fabrications, extrinsic fabrications, and false negatives. Among these, false negatives (51 % of total errors, n = 87) were the most prevalent, which predominantly occurred in noisy datasets. Large language models are particularly sensitive to input noise, which can impede their ability to identify and extract relevant information [14]. In such a case, the model often generates null values rather than incorrect predictions. This behavior reflects a mechanism designed to minimize fabrications under uncertainty, but it also underscores the model's vulnerability in handling real-world noisy data. Intrinsic fabrications comprised the second most frequent error (22 % of total errors, n = 87). Analysis revealed these errors were associated with cases where the model appeared to generate content beyond what could be directly inferred from the input data, suggesting internal information synthesis errors rather than external data inconsistencies. In contrast, extrinsic fabrications were the least prevalent error type (14 % of total errors, n = 87), indicating relatively consistent alignment between model outputs and the provided training data and contextual information. However, detailed error analysis revealed that both intrinsic and extrinsic fabrications occurred disproportionately in cases involving clinical entities with limited representation in either the RAG knowledge base or the input prompts. For example, when the task was to extract symptoms of agitation, a nursing note might mention "anxious", a relevant symptom. However, if this term was not included in the RAG knowledge file or the prompt examples, the model often failed to identify it as a symptom of agitation in dementia. Based on these empirical observations, we propose that three interventions may improve clinical LLM performance: (1) enhanced noise tolerance mechanisms, (2) expanded domain-specific knowledge bases within the RAG framework, and (3) systematic prompt optimization. Further controlled studies are needed to validate these hypotheses.

4.2. Answers to RQ2: which type of LLM, general-purpose or health-specific, performs better in clinical information extraction tasks when evaluated using holistic metrics like robustness, fairness, and contextual relevance?

To answer this research question, we compared general-purpose and health-specific LLMs on NER and summarization tasks.

4.2.1. Results

No statistically significant performance differences were found between general-purpose and health-specific LLMs. However, results suggest that health-specific LLMs achieve slightly higher accuracy, F1 score, robustness, and relevance in both NER and summarization tasks (see Supplementary Table 3). No consistent pattern was observed concerning bias and fairness.

Among the health-specific LLMs, Me-LLaMA achieved the highest performance in NER, with an accuracy of 75.18 % (±0.04 %), an F1 score of 75.00 % (±0.04 %), and a robustness score of 6.00 % (±0.32 %). PMC LLaMA followed closely with an accuracy of 74.18 % (±0.04 %), an F1 score of 74.00 % (±0.04 %), and a robustness score of 7.32 % (±0.38 %). The lowest performance in NER was observed for Asclepius (7B), which achieved an accuracy of 43.22 % (±0.15 %), an F1 score of 44.58 % (±0.16 %), and a robustness score of 16.25 % (±0.23 %). In the summarization task, Me-LLaMA once again stood out as the top-performing health-specific LLM, achieving an F1 score of 79.24 % (±0.04 %), a robustness of 4.90 % (±0.31 %), and a relevance of 78.25 % (±0.04 %). PMC-LLaMA closely followed, recording an F1 score of 77.26 % (±0.04 %), a robustness of 5.35 % (±0.38 %), and a relevance of 77.54 % (±0.04 %). The lowest performance in summarization was observed for MedAlpaca (7B), which achieved an F1 score of 47.02 % (±0.13 %), a robustness of 16.31 % (±0.26 %), and a relevance of 40.00 % (±0.14 %). In terms of fairness and bias, all three models exhibited similar results, specifically, Me-LLaMA and PMC LLaMA achieved fairness scores of 99.87 % (±0.03 %) and bias scores of 0.12 % (±0.04 %), while MedAlpaca recorded a fairness score of 99.85 % (±0.02 %) and a bias score of 0.14 % (±0.03 %).

4.2.2. Discussions

Although domain specialization can lead to slight model improvements in accuracy, F1 score, robustness, and relevance for healthcare-specific tasks such as NER and summarization, these gains are not statistically significant. One possible explanation is that health-specific LLMs, which are fine-tuned on domain-specific datasets, may show an improved understanding of healthcare terminology and concepts, thus performing better in tasks requiring in-depth clinical language comprehension and generation. This trend was observed across various metrics, particularly for tasks closely tied to the nuances of clinical practice. Among the health-specific LLMs, Me-LLaMA emerged as the top performer, potentially due to its training on the largest and most comprehensive biomedical and clinical datasets [42]. This extensive domain-centric pretraining likely enables closer alignment with the complexity and variability of real clinical settings, thereby improving its capacity to handle intricate healthcare-specific language effectively. Nonetheless, many of the datasets used to fine-tune the health-specific models consist of synthetic data or publicly available clinical data, which may not capture the full breadth and variability of real-world clinical environments [92]. This limitation may account for the modest nature of the observed improvements. Both general-purpose and health-specific LLMs displayed similar performance in terms of low bias and high fairness. This consistency is likely due to the foundational pretraining of LLMs on diverse and large-scale datasets, which maintains support for the generation of more equitable outputs.

4.3. Answers to RQ3: which RAG framework delivers the best performance in clinical information extraction tasks when evaluated using holistic metrics like robustness, fairness, and contextual relevance?

To answer RQ3, we compared LangChain and LlamaIndex RAG frameworks on NER and summarization tasks.

4.3.1. Results

Statistically significant performance gains were observed when combining the LangChain + LlamaIndex framework and LangChain alone in NER (see Fig. 3-a) and summarization tasks (see Fig. 3-b), across metrics including accuracy, F1 score, robustness, and relevance.

However, no statistically significant differences were found between the LlamaIndex framework and the combined LangChain + LlamaIndex framework on these same metrics for both tasks. Despite the lack of statistical significance, the combined framework recorded higher performance levels in both tasks, as summarized in Table 10. It is important to note that for the robustness metric (as discussed in Section 2.2.3), a

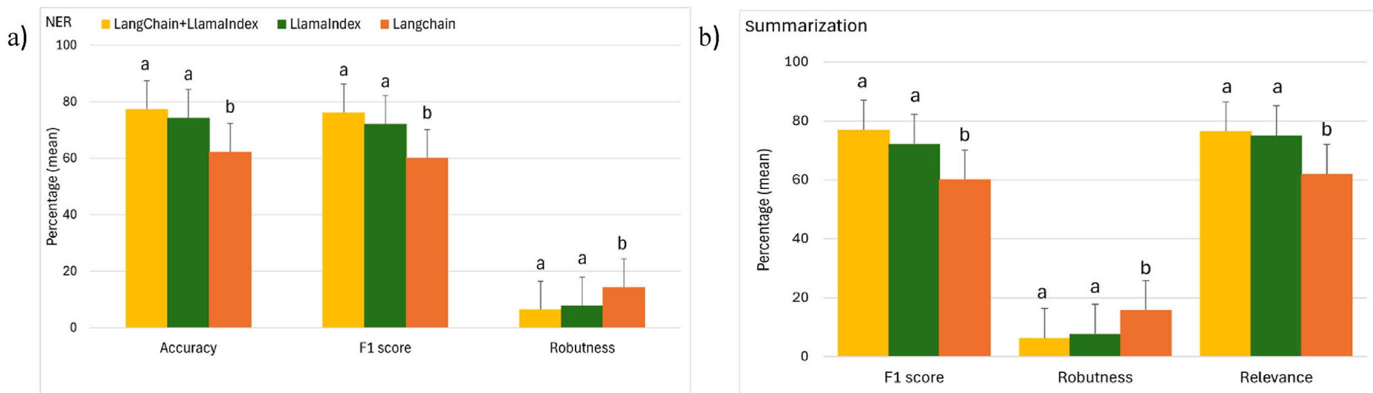

Fig. 3. Comparison of the LLMs' performance for three RAG frameworks in (a) the NER task and (b) the summarization task. Note: 'NER' denotes named entity recognition. The same notation applies to the figures that follow.

lower score indicates greater resistance to input perturbations, i.e., better robustness. In this context, the combined LangChain + LlamaIndex framework exhibits better robustness than LlamaIndex alone.

For fairness and bias, no statistically significant differences were noted among the three frameworks in either the NER or summarization tasks. Nevertheless, the combined LangChain + LlamaIndex framework exhibited marginally better performance with minimal variance across both tasks. Specifically, in NER, it achieved the highest fairness score of 99.86 % (±0.03 %) and the lowest bias score of 0.13 % (±0.01 %), respectively. In the summarization task, it recorded the highest fairness value of 99.87 % (±0.03 %) and the lowest bias value of 0.14 % (±0.02 %), respectively.

4.3.2. Discussion

Our findings suggest that combining LangChain and LlamaIndex frameworks provides improved performance in clinical IE tasks, specifically in terms of accuracy, F1 score, robustness, and relevance, than either framework used. While the improvement over LlamaIndex alone was modest, the combined framework displayed a statistically significant advantage over LangChain. This suggests that the integration of the complementary strengths of both frameworks can boost overall model performance, leveraging LangChain's workflow management capabilities and LlamaIndex's specialized retrieval algorithms [93] algorithms.

Additionally, LlamaIndex on its own outperforms LangChain on these same metrics, likely due to its more specialized design for large-scale information retrieval. By contrast, LangChain is primarily optimized for orchestrating complex workflows and context-aware interactions, which are advantageous for chatbot-like applications [93–96]. However, this strength may also represent a limitation for IE tasks, where LlamaIndex's streamlined and retrieval-focused design offers a more efficient and targeted approach. Despite these performance differences, fairness and bias metrics remained relatively consistent across the three RAG frameworks. This consistency may stem from the fact that all rely on the same underlying pre-trained models. As a result, the inherent characteristics of bias and fairness in these models are likely to remain unaffected by the specific RAG framework employed. Nevertheless, the combined LangChain + LlamaIndex framework had a modest, though not statistically significant, edge in fairness and bias scores. This marginal improvement might be due to the enhanced retrieval capabilities that better filter and rank relevant data, but its magnitude remains minor compared to the statistically significant gains observed for the primary performance metrics.

Table 10
Performance comparison between LangChain + LlamaIndex vs LlamaIndex.

Task	Metric	LangChain + LlamaIndex	LlamaIndex
NER	Accuracy	88.66 % (±0.02 %)	84.33 % (±0.03 %)
	F1 score	87.04 % (±0.03 %)	83.32 % (±0.03 %)
	Robustness	4.00 % (±0.37 %)	5.00 % (±0.41 %)
Summarization	F1 score	87.18 % (±0.02 %)	85.34 % (±0.02 %)
	Robustness	3.82 % (±0.37 %)	5.28 % (±0.38 %)
	Relevance	83.15 % (±0.03 %)	80.11 % (±0.01 %)

4.4. Answers to RQ4: what is the optimal number of examples in few-shot learning with RAG to maximize performance in clinical information extraction tasks when evaluated using holistic metrics like robustness, fairness, and contextual relevance?

To address RQ4, we evaluated the performance of one-shot, three-shot, four-shot, and five-shot examples in combination with RAG on NER and summarization tasks.

4.4.1. Results

Statistically significant performance improvements were observed when comparing one-shot learning to three-shot, four-shot, and five-shot learning strategies with RAG for accuracy, F1 score, and relevance in both the NER (see Fig. 4-a–4-b) and summarization tasks (see Fig. 4-c–4-d), for all evaluated LLMs. In contrast, no statistically significant differences were observed for robustness, fairness, or bias, although the three-shot, four-shot, and five-shot strategies exhibited modestly better performance in these metrics compared to the one-shot learning approach.

Within the NER task, no statistically significant differences were detected among the three-, four-, and five-shot strategies for accuracy, F1 score, robustness, bias, or fairness. However, a clear non-significant trend indicated that the five-shot approach achieved higher robustness than the other configurations. For instance, robustness scores for the three-shot learning for NER included 12.98 % (±0.25 %) for LLaMA-2-Chat-7B, 8.42 % (±0.21 %) for MedAlpaca-7B, 10.19 % (±0.31 %) for Baize-Healthcare, 9.14 % (±0.34 %) for ChatDoctor, 10.86 % (±0.31 %) for Asclepius-7B, 7.87 % (±0.37 %) for Mistral, 7.41 % (±0.42 %) for Gemini, 11.215 % (±0.35 %) for LLaMA-2-Chat-13B, 8.02 % (±0.42 %) for MedAlpaca-13B, 8.31 % (±0.41 %) for Asclepius-13B, 7.21 % (±0.47 %) for PMC LLaMA, 6.91 % (±0.38 %) for Me-LLaMA, 5.01 % (±0.41 %) for LLaMA-3-8B, 4.99 % (±0.42 %) for LLaMA3.1–8B.

These metrics were better than those for the five-shot learning for the same tasks. These are: 11.38 % (±0.22 %) for LLaMA-2-Chat-7B, 7.01 % (±0.21 %) for MedAlpaca-7B, 9.69 % (±0.28 %) for Baize-Healthcare, 8.04 % (±0.26 %) for ChatDoctor, 9.76 % (±0.21 %) for Asclepius-7B, 6.72 % (±0.24 %) for Mistral, 6.21 % (±0.35 %) for Gemini, 9.45 % (±0.27 %) for LLaMA-2-Chat-13B, 7.01 % (±0.32 %) for MedAlpaca-13B, 7.24 % (±0.33 %) for Asclepius-13B, 6.45 % (±0.36 %) for PMC LLaMA, 6.00 % (±0.31 %) for Me-LLaMA, 4.12 % (±0.38 %) for LLaMA-3-8B, 4.00 % (±0.37 %) for LLaMA3.1–8B

Similarly, in the summarization task, no statistically significant

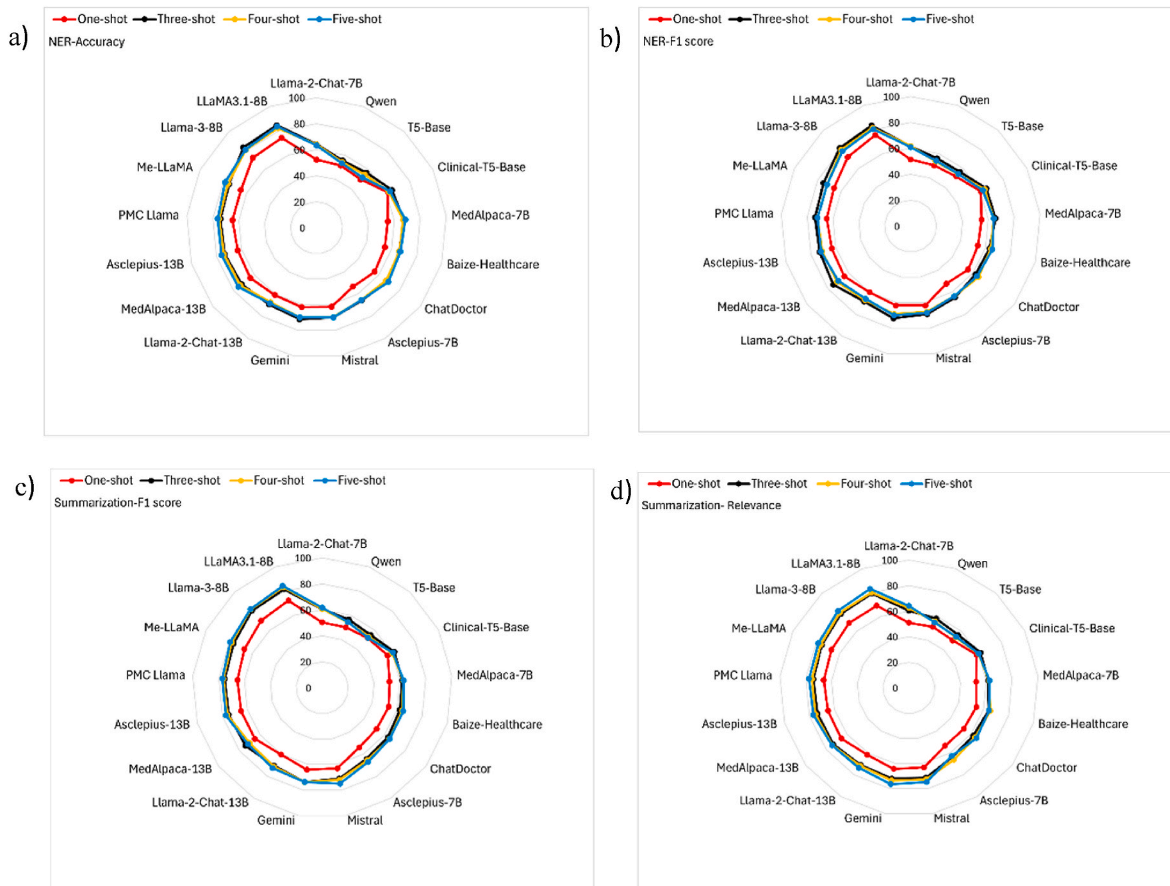

Fig. 4. Comparative evaluation of one-shot, three-shot, four-shot, and five-shot learning strategies for the NER task in (a) accuracy, (b) F1 score, and for the summarization task in (c) F1 score, and (d) relevance across the RAG frameworks.

differences were observed among the three-shot, four-shot, and five-shot strategies for F1 score, robustness, bias, fairness, or relevance. Nevertheless, a consistent non-significant trend indicated that the five-shot strategy achieved higher F1 score, robustness, and relevance than the three- and four-shot methods for all LLMs (see Table 11).

4.4.2. Discussion

While our results do not show statistically significant improvements across all few-shot configurations, accuracy and F1 scores improve up to the three-shot setting and then begin to plateau for NER. This suggests diminishing returns from additional examples beyond this point. This plateau may reflect the unique characteristics and demands of the NER task. Named entity recognition involves identifying and categorizing a relatively fixed set of entity types, which constrains the output space [97]. Since the model is provided with entity labels either through prompts or via RAG files, it may quickly learn the key patterns. Given the repetitive and formulaic nature of language in NER, once the model has seen a few representative examples of each entity, it can likely generalize effectively. Consequently, additional examples beyond this threshold may provide redundant information, especially if the examples lack diversity, resulting in limited improvements in metrics such as accuracy and F1 score. This pattern suggests that the structured nature of NER may reduce the benefit of increasing a few-shot examples, although further empirical investigation would be needed to confirm this.

Summarization tasks showed improvements across all evaluation metrics, with the five-shot configuration performing better than other settings; however, these differences did not reach statistical significance. This difference from NER tasks likely arises because summarization requires coherent, context-aware text generation. Consequently, each additional example in summarization provides new contextual cues, improving the model's understanding and enabling more accurate and comprehensive outputs. Robustness in both tasks reached its highest values in the five-shot setting, possibly due to the model's improved ability to generalize with more varied data. This enhanced generalization may help the model perform more consistently across diverse scenarios, contributing to greater robustness.

Regarding fairness and bias under few-shot learning with RAG, our findings indicate only minimal improvements as the number of examples increases. These modest changes likely reflect the fact that an LLM's inherent bias is largely determined during its original training [98]. While few-shot learning effectively refines task-specific performance, it does not substantially alter the model's foundational fairness or biases. Nonetheless, three-shot, four-shot, and five-shot learning showed a slight but consistent advantage over one-shot learning configurations. We hypothesize that this improvement can be attributed to the richer context provided by multiple examples, which helps the model interpret subtle task nuances more accurately and reduces the likelihood of biased outputs. Moreover, including more examples in the prompt may provide a more balanced representation of clinical contexts, further mitigating task-specific bias.

4.5. Study limitations and future works

In addition to the limitations discussed in Section 4.1.2 regarding the measurement of robustness, fairness, bias, and relevance, this study has a few other key limitations. The absence of certain domain-specific knowledge in the RAG files and prompts has impacted the model's ability to generate highly accurate and contextually relevant outputs. Future research should focus on expanding the scope of domain-specific

Table 11
Average performance of five-shot learning on the summarization task, evaluated across metrics including F1 score, robustness, and relevance.

Type of LLMs	LLMs	F1 score	Robustness	Relevance
General LLMs	LLaMA-2-Chat-7B	61.80 % (±0.05 %)	9.85 % (±0.22 %)	64.22 % (±0.05 %)
	LLaMA-2-Chat-13B	72.66 % (±0.04 %)	8.80 % (±0.24 %)	73.56 % (±0.04 %)
	LLaMA-3-8B	82.00 % (±0.02 %)	4.31 % (±0.30 %)	81.50 % (±0.04 %)
	LLaMA3.1–8B	**84.18 % (±0.02 %)**	**3.82 % (±0.47 %)**	**83.15 % (±0.03 %)**
	Mistral	74.99 % (±0.04 %)	8.12 % (±0.27 %)	74.88 % (±0.06 %)
	Gemini	73.77 % (±0.04 %)	6.82 % (±0.25 %)	76.58 % (±0.06 %)
	Qwen	65.82 % (±0.06 %)	8.22 % (±0.34 %)	61.88 % (±0.05 %)
	T5-Base	62.88 % (±0.06 %)	7.82 % (±0.34 %)	63.82 % (±0.05 %)
Health-specific LLMs	MedAlpaca-7B	63.00 % (±0.05 %)	9.72 % (±0.21 %)	62.58 % (±0.05 %)
	Clinical-T5-Base	63.00 % (±0.05 %)	9.72 % (±0.21 %)	62.58 % (±0.05 %)
	MedAlpaca-13B	67.78 % (±0.08 %)	6.01 % (±0.28 %)	71.21 % (±0.05 %)
	Baize-Healthcare	71.83 % (±0.04 %)	5.20 % (±0.21 %)	74.55 % (±0.04 %)
	ChatDoctor	64.91 % (±0.06 %)	8.70 % (±0.21 %)	64.33 % (±0.06 %)
	Asclepius-7B	65.01 % (±0.06 %)	8.62 % (±0.28 %)	65.21 % (±0.06 %)
	Asclepius-13B	66.99 % (±0.05 %)	8.12 % (±0.28 %)	62.58 % (±0.05 %)
	PMC LLaMA	77.22 % (±0.05 %)	5.30 % (±0.32 %)	76.85 % (±0.04 %)
	Me-LLaMA	**77.26 % (±0.05 %)**	**5.35 % (±0.38 %)**	**77.54 % (±0.04 %)**

Finally, no statistically significant differences and no clear trends were observed for fairness and bias across the three-shot, four-shot, and five-shot strategies for all LLMs in either the NER or summarization task.

training data within the RAG framework and refining the prompts, which would help enhance model performance. Additionally, the evaluation datasets, particularly for robustness, fairness, and bias, were extremely small, limiting the statistical strength of the findings. To improve the reliability of such evaluations, future studies should aim to scale up test samples, targeting at least 1000 clinical notes per metric. It is imperative that future research focuses on broadening evaluation metrics and integrating additional dimensions such as noise tolerance, interpretability, and contextual appropriateness. Equally important is employing high-quality, domain-specific clinical data in the training phase to more accurately reflect the complexities of real-world clinical environments. Investigating optimal few-shot learning configurations tailored to distinct clinical IE tasks could further enhance the models' capacity to utilize additional context effectively. Also, we recognize that while prompt standardization mitigates variability to some extent, randomness in example selection cannot be fully eliminated. As part of future work, we plan to explore automatic example sampling techniques to further reduce variability and improve reproducibility. Moreover, continued refinement of RAG frameworks, with an emphasis on enhancing the synergy between retrieval and generation components, will be essential to maximize the extraction of clinically relevant information. By addressing these recommendations, future efforts will advance the safe, effective, and ethically sound deployment of generative AI in clinical IE, ultimately contributing to improved patient care outcomes.

5. Conclusion

This study presents a comprehensive evaluation framework for clinical IE tasks, specifically designed for RAC settings and applied to unstructured nursing notes. By extending beyond conventional metrics such as accuracy and F1 score, the framework incorporates robustness, fairness, bias, and contextual relevance factors essential for real-world deployment in sensitive healthcare environments. Our results demonstrate that while LLMs achieve strong performance on traditional metrics and fairness measures, challenges persist in achieving robustness and maintaining contextual relevance. These findings emphasize the need for evaluation strategies that reflect the nuanced nature of the clinical text and the risks associated with deploying LLMs in practice. Crucially, we highlight the importance of ethical deployment, acknowledging potential failure cases and the harms that incorrect outputs could cause in clinical workflows. This underscores the need for continuous error analysis, human oversight, and responsible integration of AI to safeguard patient safety and trust.

Beyond empirical comparisons, this study contributes a generalizable framework for evaluating LLMs in clinical NLP, one that is adaptable across tasks, model types, and domains involving unstructured clinical data. It promotes a shift from narrow performance assessments to a more comprehensive evaluation paradigm that supports the safe, fair, and context-aware integration of LLMs into clinical workflows. In addition, we identified the most effective combinations of LLM types and RAG configurations for NER and summarization tasks. These insights not only inform model selection and prompt design in RAC applications but also offer guidance for future research and development of clinical AI systems prioritizing fairness, reliability, and real-world relevance.

CRediT authorship contribution statement

Dinithi Vithanage: Writing – original draft, Methodology, Conceptualization. **Ping Yu:** Writing – review & editing, Supervision, Conceptualization. **Qianqian Xie:** Writing – review & editing, Conceptualization. **Hua Xu:** Writing – review & editing. **Lei Wang:** Writing – review & editing, Supervision. **Chao Deng:** Writing – review & editing, Supervision.

Availability of data and materials

Data is not available to comply with the Australian laws on privacy protection.

Consent for publication

All authors have approved the manuscript and agree with its submission to the Journal of Computers in Biology and Medicine.

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Declaration of competing interest

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Appendix A. Supplementary data

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References

[1] S. Sivarajkumar, et al., Clinical information retrieval: a literature review, J. Healthc. Info. Res. (2024) 1–40.
[2] D. Vithanage, et al., Extracting symptoms of agitation in dementia from free-text nursing notes using advanced natural language processing, in: MEDINFO 2023—The Future is Accessible, IOS Press, 2024, pp. 700–704.
[3] Y. Zhu, et al., Agitation prevalence in people with dementia in australian residential aged care facilities: findings from machine learning of electronic health records, J. Gerontol. Nurs. 48 (4) (2022) 57–64.
[4] G. Fette, et al., Information extraction from unstructured electronic health records and integration into a data warehouse, in: INFORMATIK 2012, Gesellschaft für Informatik eV, 2012.
[5] M.Y. Landolsi, L. Hlaoua, L. Ben Romdhane, Information extraction from electronic medical documents: state of the art and future research directions, Knowl. Inf. Syst. 65 (2) (2023) 463–516.
[6] E. Hossain, et al., Natural language processing in electronic health records in relation to healthcare decision-making: a systematic review, Comput. Biol. Med. 155 (2023) 106649.
[7] D. Vithanage, et al., Contextual word embedding for biomedical knowledge extraction: a rapid review and case Study, J. Healthc. Info. Res. (2024) 1–22.
[8] D. Vithanage, et al., Adapting generative large language models for information extraction from unstructured electronic health records in residential aged care: a comparative analysis of training approaches, J. Healthc. Info. Res. 9 (2025) 191–219.
[9] M. Alkhalaf, et al., Applying generative AI with retrieval augmented generation to summarize and extract key clinical information from electronic health records, J. Biomed. Inf. (2024) 104662.
[10] P. Yu, et al., Leveraging generative AI and large language models: a comprehensive roadmap for healthcare integration, in: Healthcare, MDPI, 2023.
[11] Zhang, Y., Q.V. Liao, and R.K.E. Bellamy, Effect of confidence and explanation on accuracy and trust calibration in AI-assisted decision making, in *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*. 2020, Association for Computing Machinery: Barcelona, Spain. p. 295–305.
[12] C. Zeng, et al., A survey on machine reading comprehension—tasks, evaluation metrics and benchmark datasets, Appl. Sci. 10 (21) (2020) 7640.
[13] T. Zhang, et al., Bertscore: Evaluating Text Generation with BERT, 2020.
[14] D. Cecchini, et al., Holistic evaluation of large language models: assessing robustness, accuracy, and toxicity for real-world applications, in: Proceedings of the 4th Workshop on Trustworthy Natural Language Processing (Trustnlp 2024), 2024.
[15] R. Bommasani, P. Liang, Holistic evaluation of language models, Ann. N. Y. Acad. Sci. (2023) 1525.
[16] I.O. Gallegos, et al., Bias and fairness in large language models: a survey, Comput. Linguist. 50 (3) (2024) 1097–1179.
[17] S. Bedi, et al., Testing and evaluation of health care applications of large language models: a systematic review, JAMA 333 (4) (2024) 319–328.
[18] P. Kanithi, et al., MEDIC: towards a Comprehensive Framework for Evaluating Llms in Clinical Applications, 2024.
[19] K. Singhal, et al., Large language models encode clinical knowledge, Nature 620 (7972) (2023) 172–180.
[20] E. Bolton, et al., Biomedlm: a 2.7 b parameter language model trained on biomedical text, arXiv preprint arXiv:2403.18421 (2024).
[21] K. Patil, Llms for AI Planning: a Study on Error Detection and Correction in PDDL Domain Models, 2024.
[22] T. Searle, et al., Estimating redundancy in clinical text, J. Biomed. Inf. 124 (2021) 103938.
[23] L. Tang, et al., Evaluating large language models on medical evidence summarization, npj Digit. Med. 6 (1) (2023) 158.
[24] J. Zhang, et al., The potential and pitfalls of using a large language model such as ChatGPT, GPT-4, or LLaMA as a clinical assistant, J. Am. Med. Inf. Assoc. 31 (9) (2024) 1884–1891.
[25] F. Shi, et al., Large language models can be easily distracted by irrelevant context, in: International Conference on Machine Learning, PMLR, 2023.
[26] J. Xu, et al., Learning to break the loop: analyzing and mitigating repetitions for neural text generation, Adv. Neural Inf. Process. Syst. 35 (2022) 3082–3095.
[27] A. Elek, et al., Evaluating the Efficacy of Perplexity Scores in Distinguishing AI-Generated and Human-Written Abstracts, Academic Radiology, 2025.
[28] K. Manikantan, S. Jaganathan, A model for diagnosing autism patients using spatial and statistical measures using rs-fMRI and sMRI by adopting graphical neural networks, Diagnostics 13 (6) (2023) 1143.
[29] Y. Li, et al., AdaNI: Adaptive noise injection to improve adversarial robustness, Comput. Vis. Image Understand. 238 (2024) 103855.
[30] Y. Nemcovsky, et al., Adversarial robustness via noise injection in smoothed models, Appl. Intell. 53 (8) (2023) 9483–9498.
[31] H. Javed, S. El-Sappagh, T. Abuhmed, Robustness in deep learning models for medical diagnostics: security and adversarial challenges towards robust AI applications, Artif. Intell. Rev. 58 (1) (2024) 12.
[32] A.N. Carey, X. Wu, The statistical fairness field guide: perspectives from social and formal sciences, AI and Ethics 3 (1) (2023) 1–23.
[33] D. Moon, S. Ahn, Metrics and algorithms for identifying and mitigating bias in AI design: a counterfactual fairness approach, 13, IEEE Access, 2025, pp. 59118–59129.
[34] K. Gonuguntla, A. Shaik, P. Agrawal, Statistical bias, in: Translational Cardiology, Elsevier, 2025, pp. 179–180.
[35] B. Dhingra, et al., Time-aware language models as temporal knowledge bases, Trans. Assoc. Computat. Ling. 10 (2022) 257–273.
[36] C.-Y. Lin, Rouge: a package for automatic evaluation of summaries, in: Text Summarization Branches out, 2004.
[37] A.M.A. Zeyad, A. Biradar, Advancements in the efficacy of Flan-T5 for abstractive text summarization: a multi-dataset evaluation using ROUGE and BERTScore, in: 2024 International Conference on Advancements in Power, Communication and Intelligent Systems (APCI), IEEE, 2024.
[38] D.A. Hameed, et al., Automatic evaluating of Russian-Arabic machine translation quality using METEOR method, in: AIP Conference Proceedings, AIP Publishing, 2022.
[39] Z. Wang, et al., Enhancing diagnostic accuracy and efficiency with GPT-4-generated structured reports: a comprehensive study, J. Med. Biol. Eng. 44 (1) (2024) 144–153.
[40] A. Lahat, et al., Assessing generative pretrained transformers (GPT) in clinical decision-making: comparative analysis of GPT-3.5 and GPT-4, J. Med. Internet Res. 26 (2024) e54571.
[41] C.H. Savage, et al., General-purpose large language models versus a domain-specific natural language processing tool for label extraction from chest radiograph reports, Am. J. Roentgenol. (2024) e2330573.
[42] Xie, Q., et al., Me-LLaMA: Medical Foundation Large Language Models for Comprehensive Text Analysis and Beyond.
[43] R. Luo, et al., BioGPT: generative pre-trained transformer for biomedical text generation and mining, Briefings Bioinf. 23 (6) (2022).
[44] E. Lehman, A. Johnson, Clinical-t5: Large Language Models Built Using Mimic Clinical Text, PhysioNet, 2023.
[45] K. Singhal, et al., Large language models encode clinical knowledge, Nature 620 (7972) (2023) 172–180.
[46] W. Zhang, J. Zhang, Hallucination mitigation for retrieval-augmented large language models: a review, Mathematics 13 (5) (2025) 856.
[47] R.K. Malviya, V. Javalkar, R. Malviya, Scalability and performance benchmarking of LangChain, LlamaIndex, and haystack for enterprise AI customer support systems, in: IJGIS Fall of 2024 Conference, The New World Foundation, 2024.
[48] X. Wan, et al., Teach better or show smarter? On instructions and exemplars in automatic prompt optimization, Adv. Neural Inf. Process. Syst. 37 (2024) 58174–58244.
[49] A. Razavi, et al., Benchmarking Prompt Sensitivity in Large Language Models, Springer Nature Switzerland, Cham, 2025.
[50] A. Kwak, et al., Classify first, and then extract: prompt chaining technique for information extraction, in: Proceedings of the Natural Legal Language Processing Workshop, 2024, 2024.
[51] C.P. Chai, Comparison of text preprocessing methods, Nat. Lang. Eng. 29 (3) (2023) 509–553.
[52] K.E. Goodman, H.Y. Paul, D.J. Morgan, AI-Generated clinical summaries require more than accuracy, JAMA 331 (8) (2024) 637–638.
[53] Y. Zhu, et al., COVID-19 and its influence on prevalence of dementia and agitation in Australian residential aged care: a comparative study, Psyc. Int. 5 (4) (2024) 642–659.
[54] Y. Zhu, et al., A five-step workflow to manually annotate unstructured data into training dataset for natural language processing, in: MEDINFO 2023—The Future is Accessible, IOS Press, 2024, pp. 109–113.
[55] M. Alkhalaf, et al., Malnutrition and its contributing factors for older people living in residential aged care facilities: insights from natural language processing of aged care records, Technol. Health Care 31 (6) (2023) 2267–2278.
[56] H. Touvron, et al., Llama 2: open foundation and fine-tuned chat models, arXiv preprint arXiv:2307.09288 (2023).
[57] A. Grattafiori, et al., The llama 3 herd of models, arXiv preprint arXiv:2407.21783 (2024).
[58] A.Q. Jiang, et al., Mistral 7B. Arxiv Preprint arXiv:2310.06825, 2023.

[59] G. Team, et al., Gemini: a family of highly capable multimodal models, arXiv preprint arXiv:2312.11805 (2023).
[60] J. Bai, et al., Qwen Technical Report. Arxiv Preprint arXiv:2309.16609, 2023.
[61] C. Raffel, et al., Exploring the limits of transfer learning with a unified text-to-text transformer, J. Mach. Learn. Res. 21 (140) (2020) 1–67.
[62] T. Han, et al., MedAlpaca–an open-source collection of medical conversational AI models and training data, arXiv preprint arXiv:2304.08247 (2023).
[63] C. Xu, et al., Baize: an open-source chat model with parameter-efficient tuning on self-chat data, arXiv preprint arXiv:2304.01196 (2023).
[64] Y. Li, et al., Chatdoctor: a medical chat model fine-tuned on a large language model meta-ai (Llama) using medical domain knowledge, Cureus 15 (6) (2023).
[65] S. Kweon, et al., Publicly shareable clinical large language model built on synthetic clinical notes, arXiv preprint arXiv:2309.00237 (2023).
[66] C. Wu, et al., PMC-LLaMA: toward building open-source language models for medicine, J. Am. Med. Inf. Assoc. (2024) ocae045.
[67] D. Deutsch, R. Dror, D. Roth, A Statistical Analysis of Summarization Evaluation Metrics Using Resampling Methods, vol. 9, Transactions of the Association for Computational Linguistics, 2021, pp. 1132–1146.
[68] M. Sanz Guerrero, Performance Assessment of Credit Risk Models with Boosting Algorithms and Transfer Learning from Large Language Models, 2023.
[69] Z. Jiang, et al., Generalized demographic parity for group fairness, in: International Conference on Learning Representations, 2022.
[70] M. Hort, et al., Bias mitigation for machine learning classifiers: a comprehensive survey, ACM J. Respon. Comput. 1 (2) (2024) 1–52.
[71] N. Mehrabi, et al., A survey on bias and fairness in machine learning, ACM Comput. Surv. 54 (6) (2021) 1–35.
[72] E. Davoodijam, M. Alambardar Meybodi, Evaluation metrics on text summarization: comprehensive survey, Knowl. Inf. Syst. 66 (12) (2024) 7717–7738.
[73] B. Babaali, Evaluating neural network-based Arabic machine translation: challenges, metrics, and solutions, in: Neural Network Advancements in the Age of AI, IGI Global Scientific Publishing, 2025, pp. 317–342.
[74] S. Lee, et al., A survey on evaluation metrics for machine translation, Mathematics 11 (4) (2023) 1006.
[75] W. Yuan, G. Neubig, P. Liu, Bartscore: evaluating generated text as text generation, Adv. Neural Inf. Process. Syst. 34 (2021) 27263–27277.
[76] R. Dudley, The Shapiro–Wilk Test for Normality, MIT Mathematics, 2023.
[77] S. Seiffert, et al., Use of logit transformation within statistical analyses of experimental results obtained as proportions: example of method validation experiments and EQA in flow cytometry, Front. Mol. Biosci. 11 (2024) 1335174.
[78] P.I. Feder, et al., Analysis of proportional data in reproductive and developmental toxicity studies: Comparison of sensitivities of logit transformation, arcsine square root transformation, and nonparametric analysis, Birth defect. res. 112 (16) (2020) 1260–1272.
[79] Y. Zhou, Y. Zhu, W.K. Wong, Statistical tests for homogeneity of variance for clinical trials and recommendations, Contemp. clin. trials comm. 33 (2023) 101119.
[80] A. Nanda, et al., Multiple comparison test by Tukey's honestly significant difference (HSD): do the confident level control type I error, Int. J. Stat. Appl. Math. 6 (1) (2021) 59–65.
[81] S.W. Lee, Methods for testing statistical differences between groups in medical research: statistical standard and guideline of life cycle committee, Life Cycle 2 (2022).
[82] M.R. Taalab, et al., Intrapocket application of tea tree oil gel in the treatment of stage 2 periodontitis, BMC Oral Health 21 (1) (2021) 239.
[83] R. Doshi, et al., Quantitative evaluation of large language models to streamline radiology report impressions: a multimodal retrospective analysis, Radiology 310 (3) (2024) e231593.
[84] N. Levi, et al., Noise injection node regularization for robust learning, CoRR (2022) 15764 abs/2210.
[85] S. Frei, N.S. Chatterji, P. Bartlett, Benign overfitting without linearity: neural network classifiers trained by gradient descent for noisy linear data, in: Conference on Learning Theory, PMLR, 2022.
[86] T.D. Jui, P. Rivas, Fairness issues, current approaches, and challenges in machine learning models, Int. J. Mach. Learn. Cybernet. (2024) 1–31.
[87] S. Varma, R. Simon, Bias in error estimation when using cross-validation for model selection, BMC Bioinf. 7 (2006) 1–8.
[88] X. He, D.G. Simpson, Lower bounds for contamination bias: globally minimax versus locally linear estimation, Ann. Stat. 21 (1) (1993) 314–337.
[89] E.A. Kadhim, M.-R. Feizi-Derakhshi, H.S. Aghdasi, Advanced text summarization model incorporating NLP techniques and feature-based scoring, IEEE Access (2025).
[90] S. Afrin, et al., Resource-efficient & effective code summarization, arXiv preprint arXiv:2502.03617 (2025).
[91] V. Kieuvongngam, B. Tan, Y. Niu, Automatic text summarization of covid-19 medical research articles using bert and gpt-2, arXiv preprint arXiv:2006.01997 (2020).
[92] D. Talby, The dangers of using synthetic patient data to build healthcare AI models. https://www.forbes.com/councils/forbestechcouncil/2023/05/26/the-dangers-of-using-synthetic-patient-data-to-build-healthcare-ai-models/, 2023.
[93] O. Afify, et al., Large language model application frameworks for domain-specific chatbots, in: International KES Conference on Human Centred Intelligent Systems, Springer, 2024.
[94] B. Auffarth, Generative AI with Langchain: Build Large Language Model (LLM) Apps with Python, Chatgpt, and Other Llms, Packt Publishing Ltd, 2023.
[95] S. Ghodratnama, M. Zakershahrak, Adapting llms for efficient, personalized information retrieval: methods and implications, in: International Conference on Service-Oriented Computing, Springer, 2023.
[96] J. Pinheiro, et al., On the construction of database interfaces based on large language models, in: WEBIST, 2023.
[97] Y. Wang, et al., Nested named entity recognition: a survey, ACM Trans. Knowl. Discov. Data 16 (6) (2022) 1–29.
[98] I.O. Gallegos, et al., Bias and fairness in large language models: a survey, Comput. Linguist. (2024) 1–79.